

# Miaomiao Dai

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## EDUCATION

### University of Pennsylvania

M.S.E. in Robotics

Sep 2026 – present

Philadelphia, PA

### University of Michigan

B.S.E. in Computer Science, Summa Cum Laude (SJTU-UMich dual-degree sequence) · GPA 3.8/4.0

Aug 2024 – May 2026

Ann Arbor, MI

- Coursework: Robot Learning (A+), Algorithmic Robotics (A), Machine Learning (A), Computer Vision (A), Data Structures & Algorithms (A)

### Shanghai Jiao Tong University

B.E. in Mechanical Engineering, Global College · GPA 3.6/4.0

Aug 2022 – Aug 2024, May 2026 – Aug 2026

Shanghai, China

## PUBLICATIONS

- [1] **Miaomiao Dai**, Zhongqiang Ren. “HOP: Fast Differential Dynamic Programming for Horizon-Optimal Trajectory Planning.” *Robotics: Science and Systems (RSS)*, 2026.
- [2] **Miaomiao Dai**, Zhongqiang Ren. “AL-HOP: Augmented-Lagrangian Horizon-Optimal Planning for Fast Constrained Trajectory Optimization.” In submission to *IEEE International Conference on Robotics and Automation (ICRA)*, 2027.

## RESEARCH EXPERIENCE

### HOP: Horizon-Optimal Trajectory Planning

Jul 2025 – Feb 2026

Advisor: Prof. Zhongqiang Ren, Shanghai Jiao Tong University · RSS 2026 talk, undergraduate first author

- Standard trajectory optimizers fix the task duration in advance and re-solve for every candidate; HOP selects the optimal horizon automatically within a single solve.
- Proposed and proved a **closed-form theory** that scores every candidate horizon in one backward sweep, cutting horizon search from  $O(N^2)$  to  $O(N)$  — up to **40×** faster than brute force, with identical selected horizon and cost.
- Designed and iterated the experimental evaluation across cartpole, Segway, and quadrotor systems.

### AL-HOP: Constrained Horizon-Optimal Planning

Mar 2026 – present

Advisor: Prof. Zhongqiang Ren, Shanghai Jiao Tong University · first-author manuscript in submission to ICRA 2027

- Extends HOP to hard state-control constraints (obstacles, actuator limits) with an augmented-Lagrangian formulation that keeps the one-sweep horizon prediction.
- Produced solutions valid under the full nonlinear dynamics on **100/100** benchmark problems across 10 dynamical systems, vs. 79, 71, and 55 for ALTRO and direct collocation (IPOPT, SNOPT); where both were valid, baselines needed  $1.6\text{--}3.1\times$  the solve time.

### Touch Without Touch: Measured-Touch Supervision for Visual Representations

Jun 2026 – present

Advisors: Prof. Hesheng Wang, IRMV Lab, Shanghai Jiao Tong University · Prof. Lin Shao, RoboScience

- Asks whether paired, *measured* touch — pressure, force, contact — can teach an RGB representation physical-interaction structure that visual self-supervision alone does not; inference stays strictly **RGB-only** at deployment.
- **Preregistered the protocol** before training: controlled comparisons isolating pairing correctness, sensor fidelity, and how touch enters the model, each with falsification criteria frozen in advance and evaluated by zero-shot human-to-robot transfer.
- **Built the training stack**: JEPA-style latent prediction with an EMA teacher over frozen visual and tactile features; unified **five** heterogeneous touch datasets (human hand pressure, tactile arrays, GelSight imagery, force-torque) into one pipeline running across a local RTX 5090 and a multi-GPU A100 server.

### Voice-Interactive Closed-Loop VLA Manipulation (B.E. thesis)

Feb 2026 – Aug 2026

Advisor: Prof. Yunlong Guo, Shanghai Jiao Tong University · **group leader**, team of 5

- **Deployed and fine-tuned** the  $\pi_{0.5}$  VLA policy on a low-cost SO-101 arm with dual-camera observations; the team’s full system reached **70%** end-to-end success (85% pick, 75% placement) with zero safety incidents.
- **Built the VLM-policy connection**: a grounding gate that verifies spoken objects are actually in view, rewrites commands into the policy’s training-style phrasing, and blocks motion when evidence is insufficient.
- **Designed the user interface** for voice interaction and task monitoring.

## HONORS & AWARDS

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|---------------------------------------------------------------------|---------------------|
| James B. Angell Scholar, University of Michigan                     | Mar 2026            |
| University Honors, University of Michigan (3 terms)                 | Dec 2024 – Dec 2025 |
| Dean's List, University of Michigan (3 terms)                       | Dec 2024 – Dec 2025 |
| Honorable Mention, Interdisciplinary Contest in Modeling (ICM)      | May 2023            |
| First Prize, 13th SJTU Liming Cup Mechanical Innovation Competition | May 2023            |
| Outstanding Volunteer, Miyuan Volunteer Team                        | Mar 2023            |

## SKILLS

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**Methods:** trajectory optimization (DDP, augmented-Lagrangian methods), self-supervised representation learning (JEPA-style latent prediction), VLA fine-tuning & real-robot deployment ( $\pi_{0.5}$ ), VLM integration

**Languages:** Python, C++, C, MATLAB

**ML & robotics:** PyTorch, Transformers, OpenCV, NumPy, SciPy, OpenAI Gym, ROS, PyBullet, Gazebo

**Tools:** Git, Linux (Ubuntu), Docker,  $\text{\LaTeX}$ , SolidWorks